

Molecular Tumor Board Recommendation September 17, 2024

Dear Dr. Maskey:

The Markey Cancer Center Molecular Tumor Board met on September 17, 2024 and reviewed your patient, Wilmer “Joe” Chambers (000256362; squamous cell carcinoma of the upper lobe of the lung). After reviewing the genomic report (Caris report dated August 9, 2024) and clinical characteristics of the case, the MTB recommends consideration of the following:

- Consider off-label afatinib or osimertinib targeting the EGFR mutation (G719A). This is a category 1 evidence level recommendation. Please note, first generation EGFR TKI's lack efficacy against uncommon EGFR mutations. In contrast, afatinib has demonstrated activity against these rare mutations, including G719A, with an ORR of 60% and duration of response of 17.1 months (PMID: 31931137). In addition, osimertinib has demonstrated activity against rare EGFR mutations, including G719X, with an ORR of 50-62.5% and PFS of ~13 months (PMID: 35832438, PMID: 31825714).
 - Yang JC, Schuler M, Popat S, Miura S, Heeke S, Park K, Märten A, Kim ES. Afatinib for the Treatment of NSCLC Harboring Uncommon EGFR Mutations: A Database of 693 Cases. J Thorac Oncol. 2020 May;15(5):803-815. doi: 10.1016/j.jtho.2019.12.126. Epub 2020 Jan 10. PMID: 31931137.
 - Eide IJZ, Stensgaard S, Helland Å, Ekman S, Mellempgaard A, Hansen KH, Cienas S, Koivunen J, Grønberg BH, Sørensen BS, Brustugun OT. Osimertinib in non-small cell lung cancer with uncommon EGFR-mutations: a post-hoc subgroup analysis with pooled data from two phase II clinical trials. Transl Lung Cancer Res. 2022 Jun;11(6):953-963. doi: 10.21037/tlcr-21-995. PMID: 35832438; PMCID: PMC9271433.
 - Cho JH, Lim SH, An HJ, Kim KH, Park KU, Kang EJ, Choi YH, Ahn MS, Lee MH, Sun JM, Lee SH, Ahn JS, Park K, Ahn MJ. Osimertinib for Patients With Non-Small-Cell Lung Cancer Harboring Uncommon EGFR Mutations: A Multicenter, Open-Label, Phase II Trial (KCSG-LU15-09). J Clin Oncol. 2020 Feb 10;38(5):488-495. doi: 10.1200/JCO.19.00931. Epub 2019 Dec 11. PMID: 31825714; PMCID: PMC7098834.
- Please note, the EGFR activating point mutation and RBM10 mutation are strongly suggestive of a lung primary and Caris GPSai supports this as well. Upon consultation with our pathologist, the morphology, staining, and molecular characteristics are consistent with a squamous cell carcinoma.
- Consider germline testing based on the patient's diagnosis of a kidney oncocytoma. This is a category 1 evidence level recommendation.

For questions concerning germline testing, please contact Justine.cooper@uky.edu

Thank-you for submitting this case to the Markey Cancer Center MTB.

Signature: _____ Date: _____

Signature: _____ Date: _____

Signature: _____ Date: _____

Level of Evidence	Interpretation	Supporting Evidence
1	FDA-labeled indication or NCCN-recommended; substantial evidence supporting use (standard of care)	Meta-analysis, RCTs, etc. OncoKB level 1
2	Off-label use; substantial evidence supporting use	Phase II studies and above OncoKB level 2 and above
3	Off-label use; low to moderate evidence supporting use; MTB recommends, clinical trial preferred	Phase I, case studies and reports OncoKB level 3 and below
4	Off-label; no evidence supporting use; enrollment in clinical trial recommended; report experience to MTB if used outside clinical trial	Little to no evidence (pre-clinical, animal/cellular models), mechanistically plausible

University of Kentucky Germline Testing Evidence Scale

Recommendation (based on color)	Evidence Level
Standard of care: NCCN-recommended and/or ESMO PMWG-recommended.	Category 1
Clinically and analytically valid: NCCN-consideration or expert consensus for testing.	Category 2
Clinically and analytically valid: Consider testing if clinical situation warrants.	Category 3